

Geochemical and Morphological Indicators of Glacial Till Age in Colorado's Upper Arkansas River Valley

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Abstract. This study aims to refine methods for identifying and relatively dating glacial till deposits within the upper Arkansas River Valley, Colorado. By collecting and analyzing till samples from moraines attributed to the Pinedale, Bull Lake, and Pre-Bull Lake glaciations, we investigated effective field and laboratory techniques for distinguishing glacial landforms and estimating their relative ages. Key methods included moraine slope measurements, organic carbon analysis, grain size distribution, and XRF geochemical analyses focused on rubidium-strontium (Rb) and zirconium-barium (Zr) ratios. The results indicate that moraine slope measurements and Rb XRF data provided the most consistent relative dating across sampling locations, while organic carbon analysis showed mixed results, likely due to sample contamination from residual organic material. Grain size and Zr analyses did not correlate with moraine age as anticipated, suggesting their limited applicability for dating in this context. These findings highlight the potential of specific geochemical and morphological methods in glacial dating and suggest improvements for future sampling and analysis approaches.

Introduction

This study explores glacial geomorphology within the upper Arkansas River Valley, Colorado, focusing on moraines formed by past glaciations. Our research team collected glacial till samples from key moraines in this region, aiming to better understand and recognize glacial till and landscape features. By analyzing the physical and chemical characteristics of these tills, we address two main research questions: (1) What distinct properties in the field and laboratory identify glacial till deposits and related landforms? (2) Which methods are effective for relatively dating glacial landforms in this area?

Previous studies have documented the glacial history of the Arkansas River Valley, attributing landforms to distinct glacial episodes such as the Pinedale, Bull Lake, and Pre-Bull Lake glaciations (Schweinsberg et al., 2020). However, field identification and dating methods for glacial features in this landscape remain challenging, as these deposits exhibit complex depositional histories and post-depositional alterations. Through a combination of geomorphological observations, soil sampling, and laboratory analysis, we hope to refine techniques for till identification and age assessment, contributing to a

broader understanding of glacial chronology and landscape evolution in the Rocky Mountains.

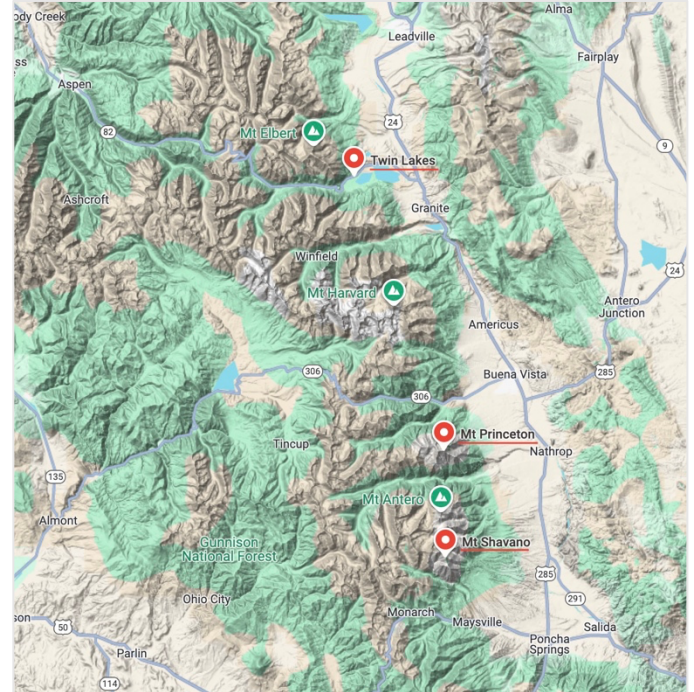


Figure 1. Terrain map of the Upper Arkansas River Valley, Colorado, with sampling locations marked in red.

Methods

Samples were collected at Twin Lakes, Mount Shavano, and Mount Princeton. The samples were taken from moraines from the Pinedale, Bull Lake, and Pre-Bull Lake glaciations. The Twin Lakes sampling locations were sparsely vegetated with shrubs and cacti. The Mount Shavano moraines were fairly vegetated with shrubs and cacti. The vegetation at Mount Princeton largely consisted of pine trees, fir trees, currant, and cacti.

Soil samples were collected by digging down between 10 and 20 cm to the A layer of soil, avoiding the organic layer, and using a hand shovel to collect till in plastic bags. One bag was collected at each site. For the cobble shape analysis, twenty cobbles and pebbles ranging from golf ball-sized to softball-sized were collected at each site.

These were then measured in the field using calipers along three axes, with “A” being the longest side, “B” the intermediate, and “C” the shortest side. The dimensions of each cobble were used to make tri-plots for analysis. Slope measurements were taken of the ice proximal side of each moraine using an inclinometer.

Soil samples were then oven-dried in their plastic bags at 70°C for 24 hours. The dried soil samples were then sealed to prevent the absorption of moisture from the air. The samples were then sieved with a 2mm sieve and flat pan. Visible roots and twigs were then removed, and the < 2 mm fractions for each sample were placed in new Ziploc bags for use in organic carbon, grain size, and XRF analyses.

For the organic carbon analysis, samples were loaded into crucibles and weighed before and after going into the muffle furnace for six hours at 500°C. Once the organic carbon had been burned off, the percent mass lost on ignition was calculated.

The grain size analysis was conducted using the Malvern Mastersizer 3000. The soil samples were dispersed in deionized water, and the machine used laser diffraction to measure particle sizes. The Mastersizer 3000 uses the intensity and pattern of light scattered by the particles to determine the percentage of each sample that was sand (particles sized 0.05 to 2.0 mm), silt (0.002 to 0.05 mm), and clay (<0.002 mm).

Again, using soil < 2 mm, samples were packed into cups and covered tightly with plastic film. Using the XRF machine, the amounts of Rubidium, Strontium, Barium, and Zirconium were measured. The Rb: St and Ba: Zr levels were then plotted for analysis.

Results

Samples taken at Twin Lakes are labeled “TL” followed by their corresponding sample number. Mount Shavano samples are labeled “MS,” and Mount Princeton Samples are labeled “MP.”

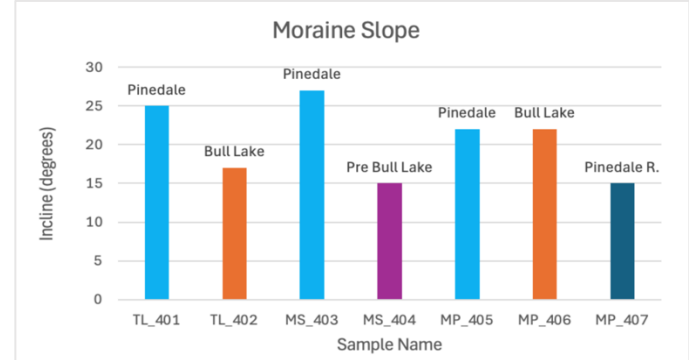


Figure 2. Bar graph of moraine slope for each of the seven sites. Measurements are sorted by sample location and then from oldest to most recent depositions.

The moraine slope data indicate that for each site, the Pinedale moraines have the steepest slopes, with Bull Lake, Pre-Bull Lake, and Pinedale Recessional moraines following.

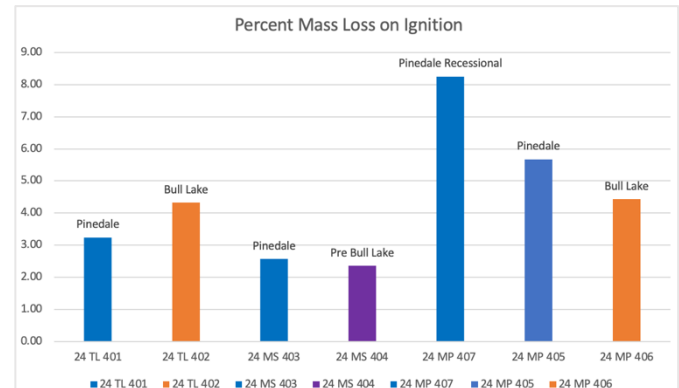


Figure 3. Bar graph of percent mass loss on ignition for each of the seven samples. Measurements are sorted by sample location and then from most recent to oldest depositions.

The percent mass loss on ignition analysis shows that the Pinedale samples lost more mass during ignition than the Bull Lake and Pre-Bull Lake samples for the Mount Princeton and Mount Shavano sites. However, the opposite was true for the Twin Lakes samples. The Mount Princeton samples had the highest percent mass loss on ignition and the Mount Shavano samples had the lowest.

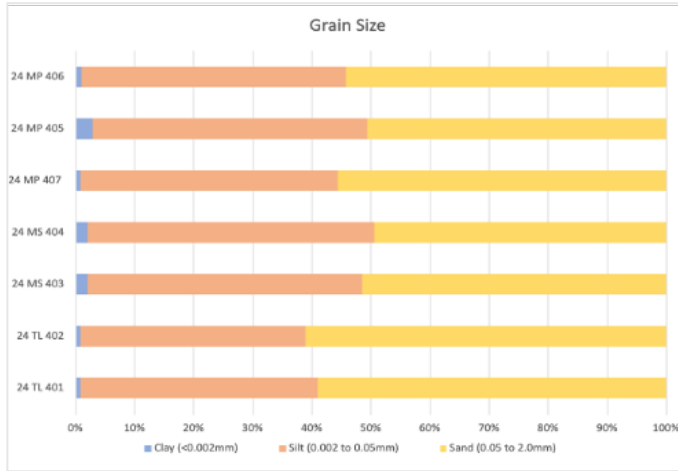


Figure 4. Bar graph of each soil sample's percentage of sand, silt, and clay, as determined by the Malvern Mastersizer 3000.

The grain size analysis using the Malvern Mastersizer 3000 showed that the Twin Lakes sites had the smallest percentage of clay particles. The Mount Shavano samples had more clay. The samples from Mount Princeton had a wide range of clay percentages with sample 24-MP-405 having the highest clay content of all the sites and samples 24-MP-406 and 24-MP-407 having some of the lowest. There was no apparent difference or trend in particle size related to the ages of the moraines.

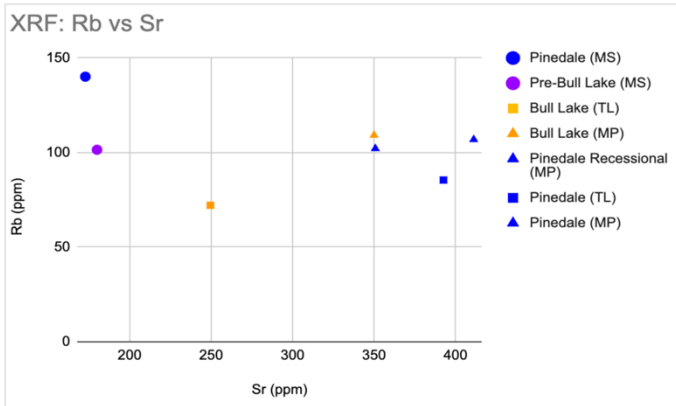


Figure 5. Rubidium and Strontium XRF data for all seven soil samples.

The XRF data for Rubidium and Strontium levels in our samples showed that the Mount Shavano sites had the lowest Sr levels and relatively high Rb levels. The Mount Princeton samples had relatively high Sr and Rb levels. Samples from Pinedale moraines tended to have higher Sr levels except for the Mount Shavano sample.

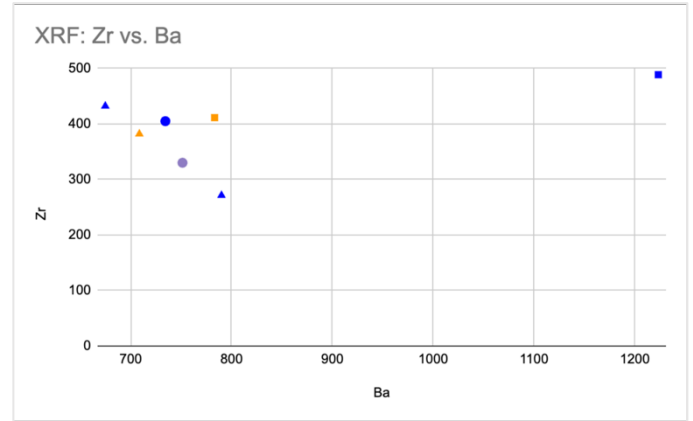


Figure 6. Zirconium and Barium XRF data for all seven soil samples.

The Zirconium: Barium XRF data show most of the Pinedale samples as having high concentrations of Zr. The Twin Lakes Pinedale glacier in particular exhibits the highest Zr and Ba levels by far.

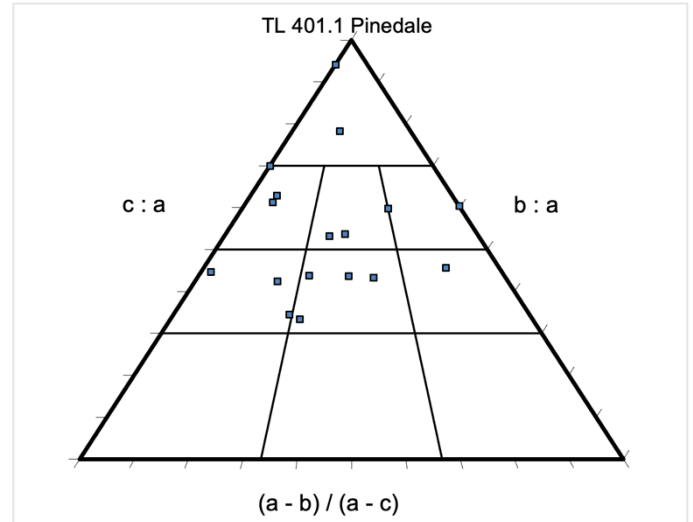


Figure 7. Tri-plot of cobble side length ratios for a Twin Lakes Pinedale moraine. For this tri-plot, cobbles that have equal A, B, and C side lengths are plotted at the top corner, long and thin cobbles are plotted in the bottom left corner, and wide flat cobbles are plotted towards the bottom right corner.

The tri-plots demonstrate that most of the samples have clast morphologies, indicating till characteristics. However, the clast shape distribution varies and there isn't a clear pattern. For example, for TL 401.1 Pinedale, the clasts are assembled in the middle-upper region of the triangle while MP 405.1 Pinedale clasts are primarily in the upper part of the plot. Interestingly, MS 404.1 Pre-Bull Lake exhibits the strongest till identity, and MP 407.1 Pinedale Recessional lies on the opposite end of the

spectrum, the clasts are widely scattered on the plot, indicating the lowest till identifying characteristics.

Discussion

The moraine slope data were similar to what was expected based on the idea that moraines become less steep as they age (Hallet and Putkonen, 1994). For all three of the sampling locations, Pinedale moraines were steeper than the older moraines. However, the Pinedale Recessional moraine at Mount Princeton contradicted this. Being a recessional moraine, it is understandable that it might behave differently than the other moraines that were sampled. Overall, moraine slope measurements appear to be an accurate method for the relative dating of moraines.

We expected the amount of organic carbon and mass lost during ignition to be higher for the older moraines because organic carbon levels tend to increase over time. However, our data only follow this trend for the Twin Lakes sites. The Mount Shavano and Mount Princeton samples had organic carbon levels that were the opposite of the expected trend. This may be due to the fact that the sampling locations for the Mount Princeton Pinedale moraines had a relatively large amount of vegetation, and some organic matter may not have been sieved out, increasing the perceived organic carbon levels in the samples. Because of this, organic carbon could still be an accurate method of dating moraines, and it would be beneficial to take more samples at each site, being more careful not to use any pine needles or roots in the analysis.

For the grain size analysis, the expected trend is that the percentage of clay particles would increase over time, meaning that the older moraines would have higher clay contents. However, the data did not follow this trend well. The clay content in the samples appears to be better differentiated by sampling location. For example, the Twin Lakes samples are both approximately 0.9% clay and the Mount Shavano samples were both approximately 2.1% clay.

Rather than being sorted by age, the XRF data for Rubidium and Strontium levels in our samples appear to be more closely grouped by sampling location. They showed that the Mount Shavano sites had the lowest Sr levels and relatively high Rb levels. The Mount Princeton samples had relatively high Sr and Rb levels. Samples from Pinedale moraines tended to have higher Sr levels except for the Mount Shavano sample. Therefore, Rb: Sr XRF data appear to be an accurate dating method for the

Twin Lakes and Mount Princeton moraines, but not the Mt. Shavano moraines.

Zr and Ba can be described as “immobile,” meaning they do not wash out of soils easily with water. They have been shown to increase over time in soils in the semi-arid west due to input from dust (Ferrier et al., 2011), so we may expect Zr and Ba to increase with age in our till samples. However, the XRF data for our sites did not comply with this expectation and, in some cases, even contradicted it. For example, three of the four Pinedale samples have some of the highest Zr concentrations, even though they are the youngest. Additionally, the Pre-Bull Lake sample (oldest) had the second-lowest Zr concentration. Overall, the Zr: Ba XRF analysis did not appear to accurately date any of the samples in this study.

The tri-plots demonstrate that most of the samples have clast morphologies indicating till characteristics. However, the clast shape distribution varies, and there is not a clear pattern in relation to the depositional age of the moraines. For example, for TL 401.1 Pinedale, the clasts are concentrated toward the middle-upper region of the triangle, while MP 405.1 Pinedale clasts are primarily in the upper part of the tri-plot. Interestingly, MS 404.1 Pre-Bull Lake exhibits the strongest till presence, and MP 407.1 Pinedale Recessional lies on the opposite end of the spectrum, the clasts are widely scattered on the plot, indicating the lowest till presence. This goes against the theory posited by Benn and Evans (2004), that younger moraines should indicate more till-like forms (closer to the apex) while older moraines should indicate more scree-like forms. It is possible that the cobbles and pebbles chosen were not a good representation of the overall population of pebbles and cobbles at the sites or the clast shape methodology needs improvement.

Conclusion

This research provided valuable insights into identifying and relatively dating glacial tills in the upper Arkansas River Valley, Colorado. The moraine slope measurements and Rb XRF data proved to be the most effective methods for estimating relative ages of the sampled moraines, aligning well with expected trends in some locations. The organic carbon analysis offered partial support for dating moraines, though discrepancies likely stemmed from unaccounted organic materials in certain samples. Grain size and Zr analyses, however, did not produce consistent age-related trends, indicating limitations in using these methods for dating in this setting.

Several aspects of the study could be improved in future work. Increasing the sample size at each site and refining sampling methods to ensure consistent removal of organic material could enhance the accuracy of organic carbon analysis. Additionally, improving the clast shape methodology or exploring alternative approaches for assessing till characteristics could clarify the morphological trends across moraines. Future research might also consider cosmogenic exposure dating to provide absolute age constraints, supplementing the relative dating methods used here.

Expanding this study could further our understanding of glacial landform evolution and refine dating techniques for glacial deposits in similar semi-arid, mountainous regions.

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